

ML4HOR: Machine Learning for Health Outcomes with R

Full Program, September-October 2026 Cohort

Imad EL BADISY, PhD

1 Course Overview

Title: Machine Learning for Health Outcomes with R

Format: Online, cohort-based

Duration: 3 weeks, 9 sessions, 2 hours per session, 18 contact hours

Dates: 28/09/2026 – 16/10/2026

Days: Monday, Wednesday, Friday (18h00 to 20h00)

Time: Morocco time

Instructor: Imad EL BADISY, PhD

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1.1 Course Description

ML4HOR is the advanced course of the MHDSR pathway: a practical introduction to machine learning for health outcomes research using R. It covers the full supervised workflow (from classic models to boosting and neural networks), unsupervised learning and clustering for patient phenotyping, honest tuning and model selection, evaluation and validation, and interpretability.

The final sessions move from analysis to a reproducible ML pipeline driven by a command-line agent, including LLM-assisted tasks such as structured extraction and text classification, with their limits and guardrails.

The course emphasizes validation, data-leakage prevention, appropriate performance metrics, calibration, interpretability, clinical relevance, and reproducible reporting.

1.2 Target Audience

Researchers, PhD students, clinicians, public health professionals, and data analysts.

1.3 Prerequisites

Basic R and statistical literacy, as covered in BIOSTATR.

1.4 Learning Objectives

By the end of ML4HOR, participants will be able to:

1. frame supervised and unsupervised problems in health outcomes research;
2. prepare tabular health datasets and avoid data leakage;
3. apply clustering for patient phenotyping and subgroup discovery;
4. fit penalized regression, tree ensembles, boosting, and multilayer perceptrons;
5. tune and select models with nested cross-validation and fair comparison;
6. evaluate discrimination, calibration, and clinical usefulness, and choose thresholds;
7. validate models internally and externally;
8. interpret models with permutation importance, PDP/ICE, and SHAP;
9. drive a reproducible ML pipeline with a command-line agent under version control;
10. use LLMs for structured extraction and text classification, and assess their outputs.

2 Detailed Schedule

Table 1: ML4HOR detailed schedule

No.	Week	Date	Time	Topic	Main Content
1	1	28/09/2026 (Mon)	18h00- 20h00	Machine learning fundamentals	Supervised vs unsupervised, bias-variance, train/test, cross-validation, resampling
2	1	30/09/2026 (Wed)	18h00- 20h00	Data preparation	Preprocessing, missing data, encoding, standardization, feature engineering, data leakage
3	1	02/10/2026 (Fri)	18h00- 20h00	Unsupervised learning and clustering	Distances, k-means, hierarchical clustering, choosing K, silhouette, stability, PCA for exploration and visualization
4	2	05/10/2026 (Mon)	18h00- 20h00	Classic supervised models	Penalized regression, decision trees, random forests
5	2	07/10/2026 (Wed)	18h00- 20h00	Boosting and neural networks	Gradient boosting, multilayer perceptron, regularization, early stopping, model comparison
6	2	09/10/2026 (Fri)	18h00- 20h00	Model tuning and selection	Hyperparameters, grid/random search, nested cross-validation, overfitting
7	3	12/10/2026 (Mon)	18h00- 20h00	Evaluation and validation	AUC, sensitivity/specificity, PPV/NPV, F1, calibration, Brier score, thresholds, internal/external validation
8	3	14/10/2026 (Wed)	18h00- 20h00	Interpretability	Variable importance, permutation importance, PDP/ICE, SHAP, local vs global interpretation

No.	Week	Date	Time	Topic	Main Content
9	3	16/10/2026 (Fri)	18h00- 20h00	Agents, LLMs and a reproducible ML pipeline	CLI agent, code generation/execution/review, versioning, structured extraction, LLM classification, limits and guardrails

3 Course Materials

- Slides and self-contained notebooks distributed before each session
- R scripts, datasets, and exercises (shared repository)
- Exercises reviewed and corrected during sessions

4 Assessment

The course ends with a short reproducible Quarto report or mini-project, presented and discussed in the final session.

5 About the Instructor

Imad EL BADISY, PhD, is a biostatistician working on statistical and machine learning methods for health data, with a focus on survival analysis. Full profile, publications, and R packages: elbadisyimad.com

6 Policies

Language: Teaching in French, with English for statistical and methodological terminology.

Software: R (≥ 4.4) and RStudio required.

Attendance: Full attendance at all 9 sessions is recommended given the cumulative structure.